

Requirements

Types of Requirements

Methods for Gathering Requirements

Choosing a Method for Gathering Needs

What are requirements?

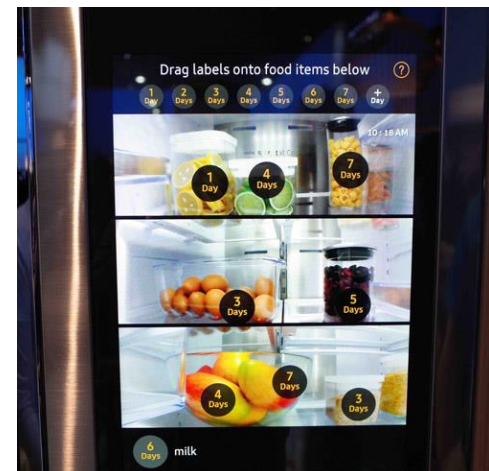
- “A requirement is a statement about an intended product that specifies what it should do or how it should perform.”
- For example, a requirement for a website might be that the time to download any file is less than 20 seconds

Requirements

- Five primary types:
 - Functional
 - Data
 - Environmental / Context of Use
 - User
 - Usability

Functional

- What the system should do
- Understanding the functional requirements for an interactive product is very important.
- Example:
 - an automated teller machine (ATM) should give users access to cash if they use their debit card
 - for a smart fridge might be that it should be able to tell when a tray is empty



Data

- What kinds/types of data need to be stored?
- How will they be stored (e.g. database)?
- Size/amount and accuracy of the data?
- Example: a banking system will need to store clients records

Environment or context of use

- The circumstances in which the interactive product will be expected to operate.
 - physical: light? noisy? vibration? dusty? heat? humidity?
 - For example, an ATM operates in a very public physical environment. Using speech to interact with the customer is therefore likely to be problematic.



Environment or context of use

- The circumstances in which the interactive product will be expected to operate.

- will data need to be shared?

If so, does the sharing have to be synchronous, e.g., does everyone need to be viewing the data at once, or asynchronous, e.g., two people authoring a report take turns editing and adding to it? do collaborators have to communicate across great distances?

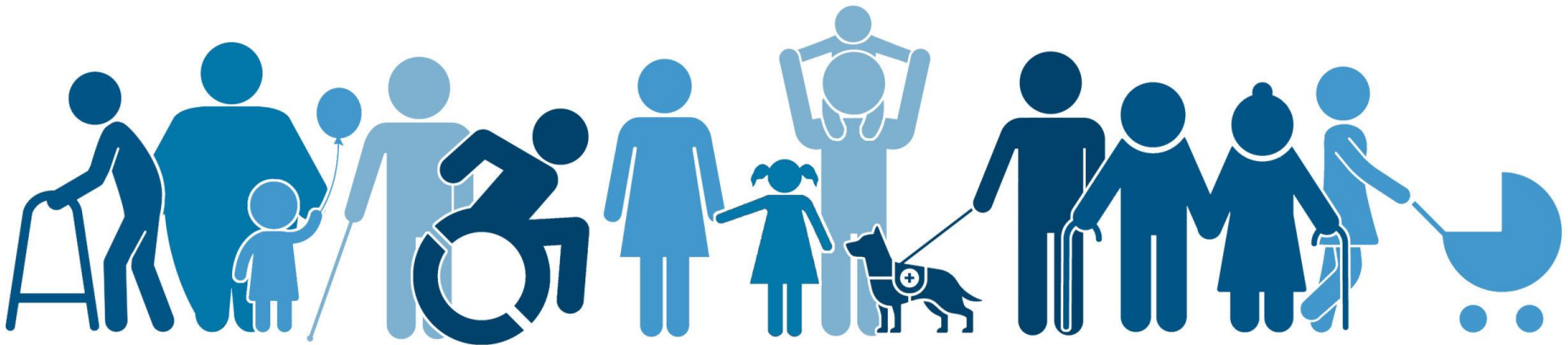


Environment or context of use

- The circumstances in which the interactive product will be expected to operate.
 - technical: what technologies will the product run on or need to be compatible with, and what technological limitations might be relevant?

Users

- User requirements capture the characteristics of the intended user group.
- User Characteristics: ability, background, attitude to computers
- Novice, expert, casual, frequent



Usability

- What is required to make the system usable?
- Should capture the usability goals and associated measures for a particular product, i.e. learnability, memorability, safety,
- Example: users should feel safe

Question

- Suggest one key functional, data, environmental, user and usability requirement for the following system:
 - A system for use in a university's self-service cafeteria that allows users to pay for their food using a credit system



Answer

- Functional:
 - the system will calculate the total cost of purchases
- Data:
 - the system must store to the price of products
- Environmental:
 - environment may be noisy and busy; users may use the system while talking
- User:
 - the majority of users will be under 30
- Usability:
 - simple to reduce wait
 - memorable for frequent users
 - deal with errors appropriately and efficiently

Question

- Suggest one key functional, data, environmental, user and usability requirement for the following system:
 - A system to support distributed design teams, e.g., for a website design team.



Answer

- Functional:
 - The system will be able to communicate information between remote sites.
- Data:
 - The system must have access to design information that will be captured in a common file format (such as AutoCAD).
- Environmental:
 - Physically distributed over a wide area. Files and other electronic media need to be shared.
- User:
 - Professional designers
- Usability:
 - Keeping transmission error rate low is likely to be of high priority

Requirement Gathering

- The purpose of requirement gathering is to collect sufficient, relevant, and appropriate data so that a set of stable requirements can be produced

Data Gathering Involving Users

- Primary methods
 - Interviews
 - Questionnaires
 - Observations
 - Focus groups

Interviews



Questionnaires



Observation



Focus Group



Image Source: https://blog.socialcops.com/academy/resources/4-data-collection-techniques-ones-right/?utm_content=59229317&utm_medium=social&utm_source=twitter

Interviews

- “Conversation with a purpose”
- Primarily for collecting qualitative data
- Can be structured, semi-structured, open-ended
 - Differ in how much control the interviewer places on the conversation

Unstructured/Open-Ended Interviews

- Resembles a conversation on a particular topic
- Also referred to as open-ended interviews
- Very exploratory in nature
 - E.g., "Can you tell me about the last time you purchased music on-line?"
- A particularly useful technique when little is known about the target domain or end users
- Despite their lack of structure, open-ended interviews should always have an objective

Open-Ended Questions



Unstructured/Open-Ended Interviews

- Strengths
 - Unstructured interviews are **more flexible** as questions can be adapted and changed depending on the respondents' answers
 - It gives the interviewer the opportunity to **probe for a deeper understanding**

- Limitations
 - It can be **time consuming** to conduct an unstructured interview and **analyze the qualitative data** (using methods such as thematic analysis).
 - Can be difficult to conduct effectively. Certain **skills may be needed** by the interviewer, the interviewer has to balance between getting the information while staying on the topic

Structured Interviews

- The interviewer asks a set of pre-planned questions in the same order for every participant
- Best for situations where you know enough about the users and domain that a list of relevant questions can be identified
 - E.g., you have an idea of what users want/need and you want to verify assumptions
- Unlike unstructured interviews, structured interviews allow for direct comparison between participants

Structured Interviews

- Often questions are closed
 - User selects an answer from a pre-determined set of alternatives
 - E.g., "Where do you use your iPhone most frequently: at home, at work, in the car?"

Structured Interviews

- Strengths
 - Easy to replicate as a **fixed set of closed questions** are used, which are easy to quantify
 - **Quick to conduct** which means that many interviews can take place within a short amount of time.

- Limitations
 - Structure interviews are **not flexible**. New questions cannot be asked impromptu (i.e. during the interview) as an interview schedule must be followed.
 - The answers from structured interviews **lack detail**. Won't know why a person behaves in a certain way.

Semi-Structured Interviews

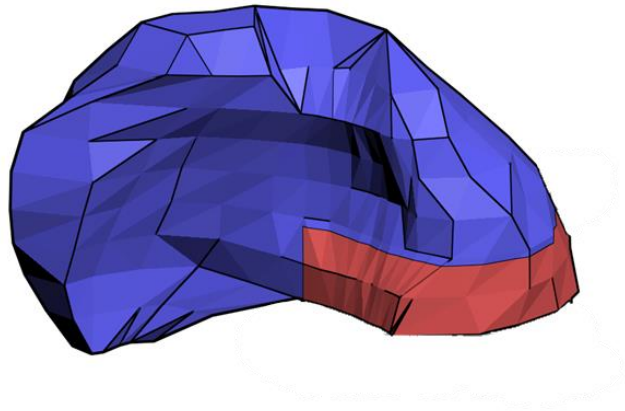
- The amount of interviewer control is in between that of an open-ended interview and a structured interview
- Interviewer has a basic set of questions, but can probe for further detail
 - E.g., “When was the last time you used your iPhone” Answer: Last night?
 - “What were you using it for?” Answer: I was using google maps
 - “For what reason?”....
- Commonly used in HCI

Semi-Structured Interviews

- Probes can take the conversation in new directions
 - Allows for some exploration
- But some comparison among participants is still possible
 - The same basic questions are asked of all participants

Interviewing Technique

- Have a **clear objective**, even when conducting an unstructured interview
- With unstructured and semi-structured interviews, ask participants to walk you through **specific examples**
- **Artifacts** can also help ground the discussion
 - ask to see examples of work that your participants have done using an application or technique
 - show them a prototype you are working on and have them comment on it

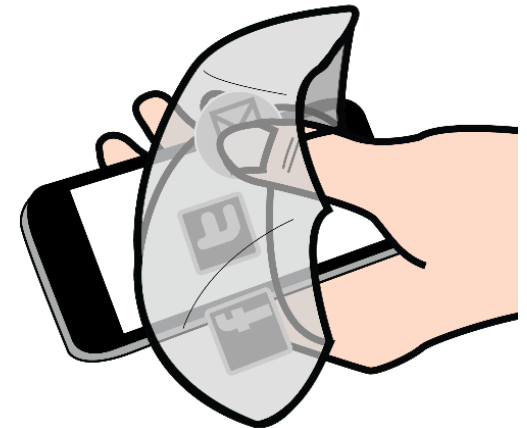
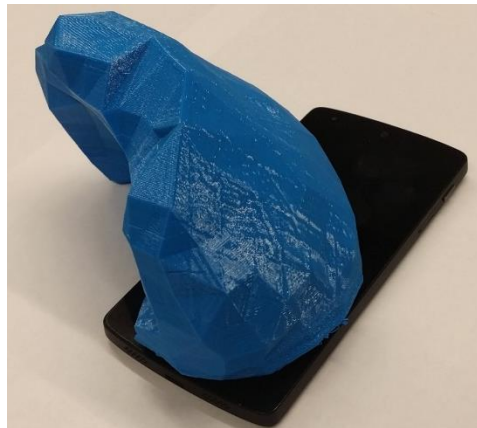
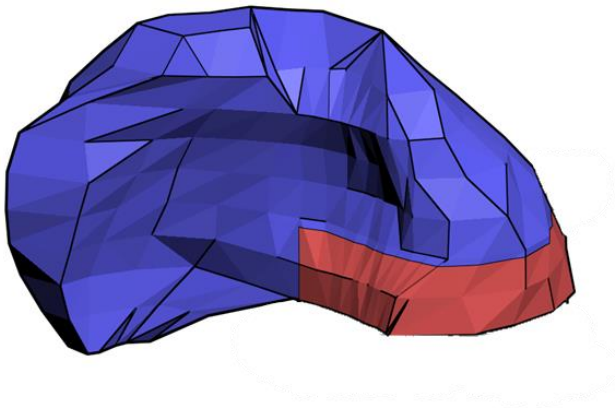




<https://www.youtube.com/watch?v=7a0e4BC5BgM>

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Interviewing Technique

- Avoid jargon
- Use the participant's terminology
- Avoid compound questions
 - Break complex questions into a series of shorter ones
- Don't interrupt
 - Give users sufficient time to answer before moving onto the next question

Sample questions: <https://usability.yale.edu/understanding-your-user/user-interviews/user-interview-example-questions>

Questions

- **Open-ended questions**

- asks for unprompted opinions
- e.g.: Can you suggest any improvements to the interfaces?

- **Closed questions**

- restricts the respondent's responses by supplying alternative answers
- e.g., Do you use computers at work: O often O sometimes O rarely

- **Ranking**

- Rank in order, please rank the following cities you like from most to least: (1) Newyork City (2) Las Vegas (3) Gainesville (4) Miami

- **Ratings**

- Normally a 5pt Likert scale is used.
- For example: 1- not at all satisfied, 2- slightly satisfied, 3-moderately satisfied, 4-very satisfied, and 5-completely satisfied

Interviews: Data Collection

- Need to record user answers
 - Otherwise there is a risk of forgetting something
- Structured interviews
 - The interviewer typically has a sheet where s/he can quickly fill in answers (e.g., questions with checkboxes for answers)
- Unstructured, semi-structured, interviews
 - Audio tape
 - Video tape
 - Notes
- **Must always** have permission to record audio/video

Interviews: Format

- A typical interview consists of the following phases
 - Introductions
 - objective, ethical issues, permission to use recording devices
 - Warm-up questions
 - easy questions to make the participant comfortable (e.g., demographic questions)
 - Main session
 - questions asked in a logical sequence
 - Cool-off period
 - easy questions if it has been a tense/difficult interview
 - Closing
 - thank the participant for his/her time

Questionnaires

- Consisting of a series of questions for the purpose of gathering information from respondents.
- Questionnaires can be thought of as a kind of written interview
- They can be carried out by telephone, computer or post.



53. What are the most frequent problems you experience with computer output?

A. _____

B. _____

C. _____

Questionnaires

- Can contain a mixed of open and closed questions
- Can collect both qualitative and quantitative data
- The design of the questionnaire is crucial; should be tested on a few people before having the target population complete it



53. What are the most frequent problems you experience with computer output?

A. _____

B. _____

C. _____

Questionnaires

Conciseness: questions should be clear and specific

e.g. should the system include a users manual? (YES/NO)

Closed questions: when possible, ask closed questions and offer a range of answers

e.g. How often do you print checks? (1: very often – 5: never)

Alternate option: Consider including a “no-opinion” option for questions that seek opinions

e.g. the payroll module is essential (...N/A)

Questionnaires

Order: think about the ordering of questions. General questions should precede specific ones

e.g. a question about a specific feature in a payroll module should come after asking whether the payroll module is essential

Instructions: Provide very clear instructions on how to complete the questionnaire

E.g., if you expect users to check a box, say so

Questionnaires

- Strengths
 - Allow researchers to **carefully structure and formulate** the data collection plan
 - Respondents can take these questionnaires at a **convenient time** and think about the answers at their own pace
 - Questionnaire can **reach every corner** of the globe
- Limitations
 - **Questionnaires without human intervention** can be quite passive and miss out on some of the finer nuances, leaving the responses open to interpretation.
 - **Response rates** can be quite low
 - Data can be noisy

Observations

- Observations can be a very **quick and effective** way of collecting data with **minimal intrusion**.
- Observe users as they are conducting a task



Coffee Shop: Expected vs Actual



Which buttons are used most?



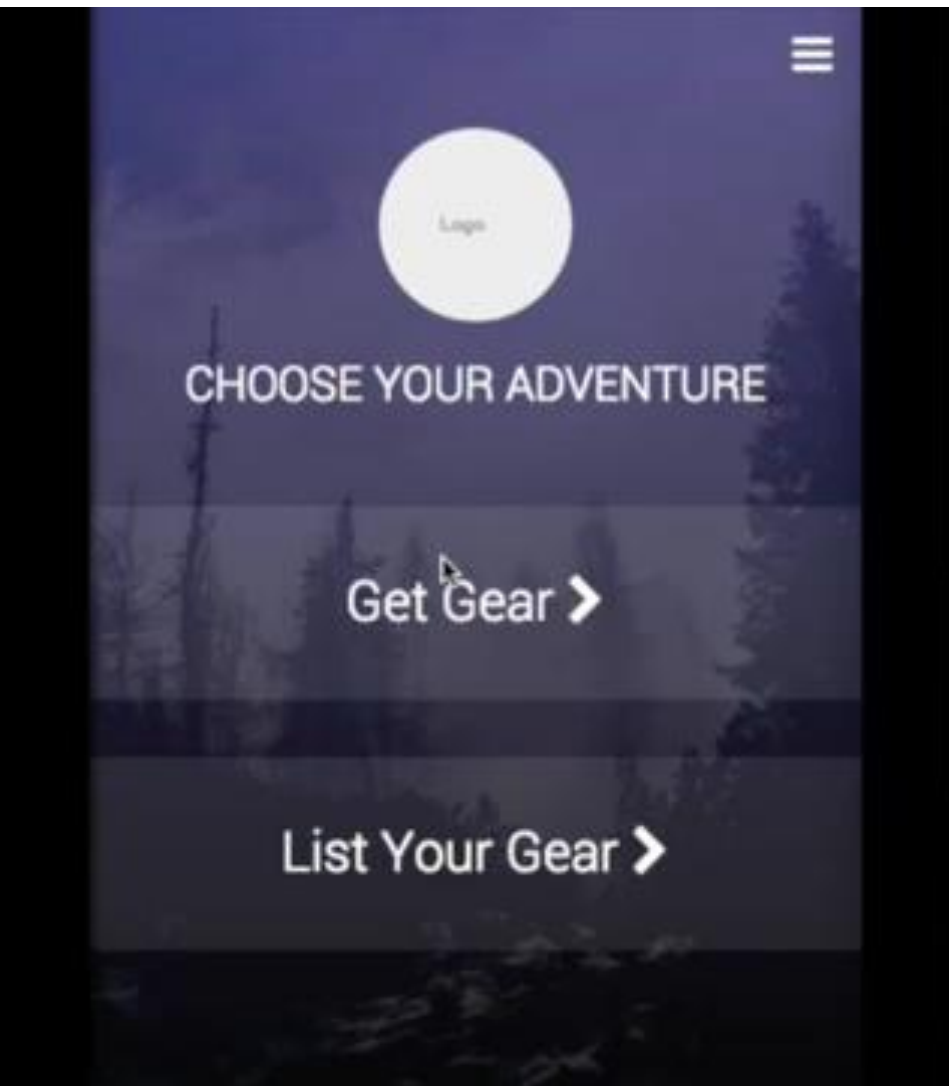
**What people say they want
vs. what they actually do**

Types of Observation

- Can observe users in the **natural environment**
- Or have users come into the **lab** to interact with prototypes
- The former is more appropriate for the initial data gathering phase

Observations

- Sometimes it is enough to simply observe the mechanics of the task
- Other times you also need to know what the user is thinking
- **Think aloud technique:**
 - Ask users to give a running commentary of what they are thinking about
 - If they forget to think aloud, remind them



Observations

- Advantages:
 - **Simple** and doesn't require **interpretation** (e.g. the number of cars driving through an intersection per hour).
 - **Infrastructure requirement and preparation time** are minimal for simple observations.
- Disadvantages:
 - More **complex observations** that ask observers to interpret something (e.g. how many cars are driving dangerously) require more complex training and are prone to bias.
 - There is the possibility of **missing out on the complete picture** due to the lack of direct interaction with sample subjects.

Focus Group

- Group interviews
 - a.k.a. focus groups
 - A group is gathered that is representative of target population
 - Interviewer acts as a facilitator



Focus Group

- Strengths
 - The presence of several relevant people together at the same time can encourage them to engage in a **healthy discussion**.
 - Any **inaccurate response** will most likely be countered by other members of the focus group.

- Limitations
 - **Finding groups of people** who are relevant to the survey and persuading them to come together for the session at the same time can be a difficult task.
 - The **presence of excessively loud members** in the focus group can subdue the opinions of those who are less vocal.

Quantitative vs qualitative data

Quantitative

- Measures of values or counts and are expressed as numbers.
- Data about numeric variables e.g. time measurements
- Easier version is counting things
- Harder version involves statistics

Qualitative

- Measures of 'types' and may be represented by a name, symbol, or a number code.
- Data about categorical variables e.g., Interviews, focus groups
- Easy version is identifying themes or topics
- Harder version is grounded theory or qualitative coding

Overview of techniques

Technique	Purpose	Data type
Interviews	Exploring issues	Mostly Qualitative
Questionnaires	Answering specific questions	Qualitative and Quantitative
Observation	Understanding context of user activity	Qualitative and Quantitative
Focus Group	Collecting Multiple Viewpoints	Mostly Qualitative

Some basic data-gathering guidelines

- Focus on **identifying the users' needs; think technology later**
- Involve **more than one representative** from each user group
- Use **a combination** of data gathering techniques
- Support the **process with props** such as prototypes and task descriptions
- Run **a pilot session**; throw away data but not the experience
- Consider carefully how to record and organize the data; video/audio/note taking, etc...

What did we cover

- Data gathering techniques → later can be used for evaluation
- Pros and cons of each technique
- When to choose a data gathering technique